



Renesas Extreme 32-bit Flash MCUs

All the real time control, signal processing, and connectivity you need...without penalty



Performance Without Sacrifice

100MHz – 165DMIPS – FPU Zero-Wait Flash – 500uA/MHz

RX600 series MCUs based on the revolutionary new RX CPU core are here. Full CPU performance is unleashed by the fast 90 nm Flash memory, feeding instructions to the CPU even at full speed. No waits, no stalls, no memory acceleration is required, just pure predictable performance for your embedded control and DSP tasks.

Key Features

RX High-performance 32-bit CPU

- 100MHz, 1.65 DMIPS/MHz
- 32-bit FPU and MAC integrated into the CPU core (no coprocessor)
- Harvard Architecture CPU with 5-stage Pipeline
- One clock per instruction, only 5-clock Interrupt Response

Zero-Wait 90 nm Flash

- No Memory Acceleration required, making very deterministic embedded control
- 100MHz, 10 nsec access, 64KB to 2MB

Low Power Consumption

- 500 uA/MHz with all peripherals on
- Under 1uA Deep Standby (RX610)

Code Efficiency

- Multiple-Length CISC Instructions
- 25% Smaller Code Size than popular 32-bit MCU architectures

Development Tools

- Full Integrated Development Environment with Renesas HEW, IAR, and GNU/Eclipse

Product Line

RX610

General Purpose
Up to 2MB Flash & 128KB SRAM for embedded control apps with high CPU processing capability & lots of I/O.

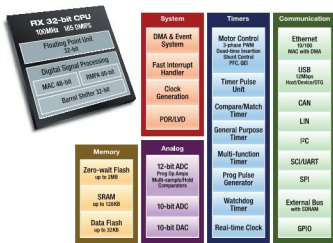
RX62N & RX621

Network
Ethernet (N/A on RX621), USB Host/Device/OTG, CAN & 12-bit ADC for networked embedded control apps. Up to 2ch USB Host/Device/OTG.

RX62T

Motor & Inverter Control
Specific timers, fast 12bit ADCs, and special op amps with programmable gain for low-cost motor control & solar panel power inverters. Reduce the number of external components. Controls up to two motors at once.

RX600 Series Block Diagram



#1 MCU
REACH
FURTHER

Experience the power of RX and learn about the entire family.

Register to be eligible to receive a FREE RX promotional board or equivalent.

<http://sg.renesas.com/ReachRX600RPB/f>



Renesas Electronics Corporation

No.1 supplier of microcontrollers in the world
Source: Gartner, "Semiconductor Applications Worldwide Annual Market Share Database" 25 March 2010. This is the 2009 ranking based on revenue.

RENESAS